MEASUREMENT LAYER · PAPER 03

Brain-computer UX is a trust-and-measurement problem

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CITATION GATE

Cleared 2026-09-04. Every item previously graded `LIKELY` was re-verified against a primary source and upgraded, including both neural-privacy bill numbers, confirmed against the legislatures' own texts. One company, Subsense Inc, could not be verified and is marked as such rather than described.

ABSTRACT

Neural-interface research optimizes the decoder. It has spent two decades improving how accurately a machine reads a brain, measured almost entirely by one metric — Information Transfer Rate. But ITR measures decoder throughput, not whether a system is usable, trustworthy, or sustainable to live with. This paper argues that the *other half* of the neural interface — how it presents its read of you back to you, how much it costs you to use, and whether you can calibrate trust in a system that infers your own internal state — is under-instrumented and under-theorized, and that it is a **measurement-and-trust problem a behavioral scientist is positioned to own, not a hardware problem.** The claim is deliberately narrow: the field is not empty — passive-BCI and neuroergonomics researchers have touched every piece of this — but no one has approached it as a first-class measurement-and-trust-calibration problem the way a behavioral scientist would. Six concrete, instrumentable open problems follow.

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1. The metric only measures one side

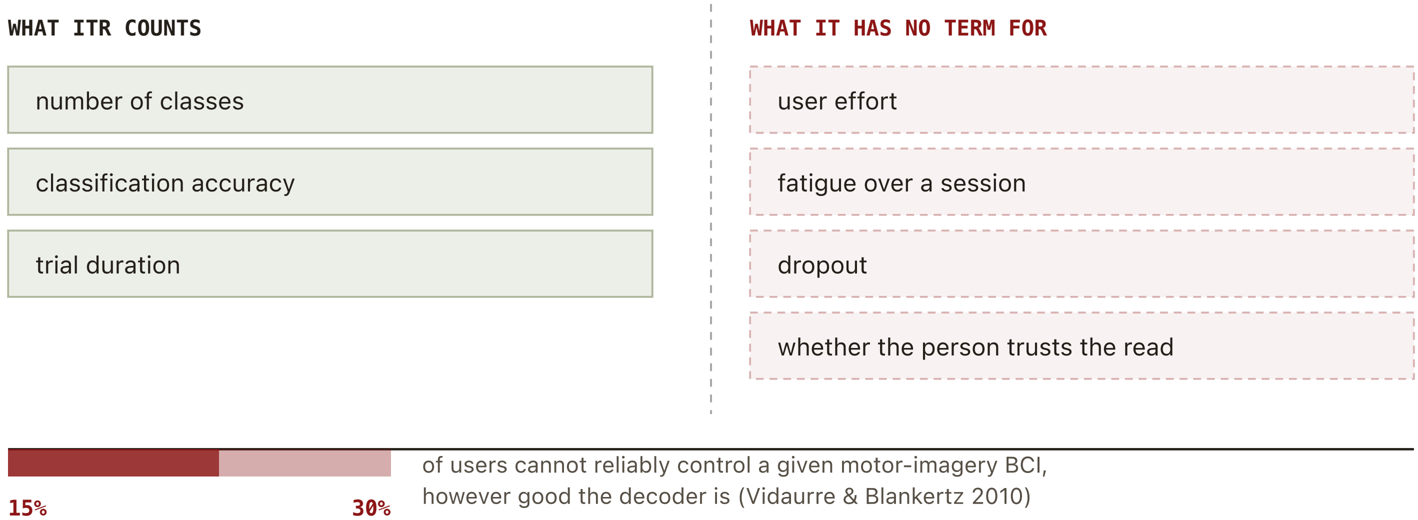

FIGURE 1 — THE METRIC AND ITS BLIND SIDE

Information Transfer Rate is built from three decoder terms and carries no term for anything the user experiences. A system can post a high number and be exhausting inside ten minutes. The band underneath is the share of people who cannot drive a motor-imagery BCI at all regardless of decoder quality, and part of that turns out to be an interface artifact rather than a biological ceiling.

The dominant performance metric is Information Transfer Rate (Wolpaw et al. 2002; bits/min from class count, accuracy, trial duration) — VERIFIED. It rewards more classes and faster trials, with **no term for user effort, fatigue, dropout, or subjective experience.** A system can post a high ITR and be exhausting to use for ten minutes.

Alongside it sits a problem hardware has not solved: **BCI "illiteracy."** Roughly 15–30% of users cannot reliably control a given motor-imagery BCI regardless of decoder quality (Vidaurre & Blankertz 2010 — VERIFIED; Allison & Neuper 2010 — VERIFIED — frame it as a population-level *design* failure, not a user failure). Crucially, part of it is an interface artifact, not a biological ceiling: Carabalona (2017, *Front. Neurosci.* — VERIFIED) shows P300 "illiteracy" is partly a stimulus-design problem. And session performance degrades as attention and fatigue degrade the very signal the system depends on — a vicious loop (Myrden & Chau 2015 — VERIFIED), instrumented from psychology (NASA-TLX) but never built into the interface itself.

The gap, stated as a measurement problem: the field has a metric for how many bits the decoder extracts and no adopted metric for what those bits cost the user. A validated "usable bits/min" or session-cost index — effort, fatigue, dropout risk, subjective burden, comparable across studies — is a concrete, citable instrument, exactly the specialty a measurement-builder brings.

2. Passive BCI inverts the human-factors problem

The product-relevant branch is passive / neuroadaptive BCI (Zander & Kothe 2011, the active/reactive/passive taxonomy — `VERIFIED`). Here the user does not intentionally drive anything; the system reads state (workload, attention, engagement, error-awareness) and acts. The mature real-world use is safety-critical operator monitoring: adaptive automation for air-traffic controllers (Di Flumeri et al. 2019 — `VERIFIED`), dry-EEG for drivers (Zander et al. 2017 — `VERIFIED`), adaptive instruction under cognitive load (Gerjets et al. 2014; Beauchemin et al. 2024 — `VERIFIED`), and the case for mental-state-based adaptive interfaces over long sessions (Hinss et al. 2022 — `VERIFIED`). This is the branch closest to funded, defensible deployment — aviation, defense, industrial — as opposed to earlier-stage consumer neurotech, and it is the branch that ties directly to human-machine teaming.

The inversion: passive systems don't need to solve intentional-control illiteracy, but they inherit a harder problem. **State inference is probabilistic and often wrong, and the system now acts on a guess about your internal state.** The human-factors question shifts from "can I drive this" to "do I trust what it just did, and did it do the right thing."

3. The readback problem: no framework for what a neural interface should show you

There is an active neurofeedback-modality literature, but it answers "which sensory channel best *trains* a target brain state," not "what information architecture, timing, and framing should govern how a system communicates its inferred read of you *back* to you." Those are different questions, and the second — the UX/trust question — is essentially unframed. Reviews categorize feedback mechanisms (Zhang et al. 2026 — `VERIFIED`) without prescribing which to choose for a given cognitive-load or attention state; the field's own researchers gesture at the gap ("more entertaining, multimodal, adaptive feedback might improve UX" — Kober, Wood & Berger 2024, `VERIFIED`) without formalizing it.

Neural interfaces carry a wrinkle no adjacent field has: **feedback about your own inferred state can change that state** — telling someone "you look distracted" measurably shifts attention and arousal (an observer effect unique to this domain). Almost no reviewed work treats this as a design constraint. Aviation alarm-design (urgency × modality × false-alarm cost) and mobile-HCI interruption research (interruption cost as a function of task state) are the analogues that exist elsewhere and do not exist for neural readback. **Status: `OPEN`.** The honest claim is not "nobody has looked" but "the pieces are touched, adjacent to their authors' specialties; no transferable design framework exists."

4. The trust problem is a cognitive-attack-surface problem pointed inward

Automation-bias research shows people mis-calibrate trust in automated agents — too much or too little (de Visser et al. 2018 — `VERIFIED`). But that research was built for systems acting on *external* data the user can independently check: radar, a thermostat. **Neural state inference removes the independent check.** When a neuroadaptive system reallocates a task, suppresses a notification, or escalates an alert because it inferred you were overloaded, you often cannot verify or contest its read of *you*. That is a categorically different trust problem than trusting a system's read of the external world, and I did not find literature that names the asymmetry and treats it as the design problem it is (`OPEN`).

This is the same structure as the Cognitive Attack Surface, turned inward: a system acting on a probabilistic model of your internal state, with the disclosure and calibration questions front and center. The countermeasures transfer — confidence display, disclosure of the inferred read, override affordances that don't themselves spike cognitive load, calibration instrumentation — and designing them is a measurement-and-trust problem, not a decoder problem. Ehrlich et al. (2023, `VERIFIED`) already show systems acting *preemptively* on anticipatory neural responses, which is where "who decided that" stops being philosophical.

5. The governance context is arriving fast

Neurorights is now a live legislative area, which makes trust and disclosure not just UX but compliance: Canada federally protected neural data (Feb 2026 — `VERIFIED` via the Neurorights Foundation); the US MIND Act was introduced Sept 2025 (`VERIFIED`); Colorado (HB24-1058, signed 2024-04-17, in force 2024-08-07) and California (SB 1223, signed 2024-09-28, Chapter 887, in force 2025-01-01) amended their privacy acts to classify neural data as sensitive (`VERIFIED` — both bill numbers confirmed against the legislatures' own texts). A system that infers internal state and cannot explain or let the user contest its read is walking into this. Trust-calibration design is becoming a regulatory surface, not only a UX nicety.

6. Six open problems a measurement specialist can own

Each is a problem a behavioral scientist — not a hardware or ML engineer — is the right kind of expert for:

1. **A usability metric beside ITR** — a validated "usable bits/min" / session-cost index (effort, fatigue, dropout, subjective burden), comparable across studies. A literal measurement-instrument contribution.

2. **A feedback-timing/modality framework for neural readback**, grounded in attention and cognitive-load theory rather than case-by-case modality trials — designed experiments manipulating interruption cost, framing, and modality against measured attention/arousal.

3. **Trust-calibration instrumentation for neuroadaptive systems** — measuring, in real time, whether a user over- or under-trusts the system's read of their own state, and designing corrections (disclosure, confidence, override) that don't spike load. The cognitive-security toolkit, pointed inward.

4. **Illiteracy as a design variable, not a user trait** — treating "can't produce a usable signal" as an interaction-design failure to test and iterate, following Carabalona's crack.

5. **Platform-scale interaction standards** for when BCI becomes an OS input method (Synchron × Apple is the leading indicator, May–Aug 2025 — `VERIFIED`) — the conventions touch and voice each needed, built by HCI researchers, not hardware engineers, and not yet built for neural input.

6. **Reward/engagement design for neurofeedback and closed-loop training** — variable-reward and engagement-without-surveillance-anxiety as first-class design variables, which the neurofeedback literature treats as a training-efficacy afterthought. Directly the reward-circuitry work of the Cognitive Attack Surface papers.

7. The honest thesis

Not "nobody has looked at neural-interface UX" — Zander/Kothe-lineage passive-BCI work, Hinss et al. (2022), and Gerjets et al. (2014) each touch pieces of it, and a credible paper cites them as partial precedent. The sharper, defensible claim: **the pieces have been touched by neuroscientists and HCI researchers working adjacent to their own specialty; no one has approached neural-interface UX as a first-class measurement-and-trust-calibration problem the way a behavioral scientist would.** That is the seat this program takes.

Note on the industry frame

Company facts (Neuralink human trials from Jan 2024; Synchron's Apple/NVIDIA OS-integration play; Precision Neuroscience's April 2025 510(k); Blackrock's company-reported metrics — treat as company claims) were live-verified this pass and date the inflection toward platform-scale neural UX. **Subsense Inc could not be verified** — no public footprint was found this pass (subsense.ai is a parked domain). It is likely an early/stealth company; confirm what it builds directly with the contact before citing it. Do not describe Subsense's product from assumption.

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All entries verified against a primary source or the issuing body on 2026-09-04.

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